

CITS3002

Computer Networks



THE UNIVERSITY OF
WESTERN
AUSTRALIA

About Me

- ❑ Dr Omid Tavallaie
 - ❑ Lecturer in Computer Science, University of Western Australia
 - ❑ Visiting Researcher, Infrastructure Group, University of Oxford
- ❑ Research Interests:
 - ❑ Distributed Systems (Federated Learning, Split Learning)
 - ❑ Computer Networks (Internet of Things, Green Networking)
- ❑ Background:
 - ❑ PhD in Computer Science (University of Sydney)
 - ❑ Postdoctoral Researcher and Lecturer
 - University of Oxford
 - University of Sydney
 - ❑ Industry experience in embedded payment systems

Teaching Team



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Timetable

❑ Lecture:

- Starts from week 1
- Friday, 12 pm – 2 pm
- School of Social Sciences, Lecture Theatre G130

❑ Labs:

- Start from week 2
- Run as drop-in sessions.
 - ❖ That means you should attend the lab when you need help.
- School of Computer Science, Lab G09
- Time:
 - Monday, 9 am – 11 am
 - Monday, 4 pm – 6 pm
 - Thursday, 8 am – 10 am
 - Friday, 8 am – 10 am

Emergency



- ❑ General emergency: call campus security at **6488 2222** (or 1800 655 222)
- ❑ In super emergency: call emergency at **000**
- ❑ In all buildings, we have an emergency procedure booklet that outlines how to react when such events happen.
 - Please take a time and read it.
- ❑ For more details, please have a read through our **emergency procedure** for various potential incidents
 - <https://www.uwa.edu.au/about/campus-services/safety/hazards-incidents-and-emergencies/emergency-procedures>

Emergency procedures

For life threatening emergencies dial 000 (if phoning from UWA dial 0 for outside line if using office phone; if using TEAMS no need to dial 0 first). Also dial UWA Security on [\(+61 8\) 6488 2222](tel:+61864882222).

Code RED - Fire/smoke	▼
Code PURPLE - Bomb threat	▼
Code BLUE - Medical emergency	▼
Code BLACK - Personal threat	▼
Code YELLOW - Internal (UWA) emergency	▼
Code BROWN - External emergency	▼
Code ORANGE - Evacuation	▼

Assessments

Assessment Item	When	Covers	Worth (total)
Mid-semester exam	Week 6 (3/4/2026) 3 pm – 5 pm	Weeks 1 to 5	20%
Group Project	Out in Week 8 Due in Week 11	Weeks 1 to 7	30%
Final Exam	During exam period	Everything	50%

Overview

Week	Lecture
1	Introduction
2	Physical Layer
3	Data Link Layer
4	Local Area Networks
5	Wireless Networks
6	Network Layer
7	Transport Layer
8	Coding and Compression

Overview

Week	Lecture
9	Client/Server Design
10	Internet Measurement
11	Networking Hot Topics
12	Review

- ☐ Weekly labs (start from week 2)
- ☐ You can do them at your own pace and get help during the drop-in sessions
- ☐ The main focus is to develop practical skills and experience implementing networking concepts

Group Project

- ☐ Given in early week 8, due end of week 11
- ☐ 30% of the total unit marks
- ☐ Each group consists of two students

Final Exam

- ☐ Worth 50%
- ☐ Covers all materials
- ☐ 120 minutes long

Commonwealth of Australia

Copyright Regulations 1969

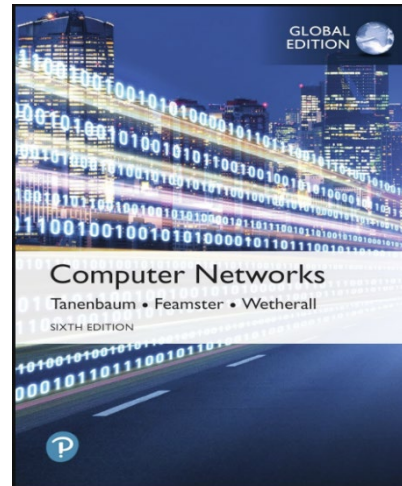
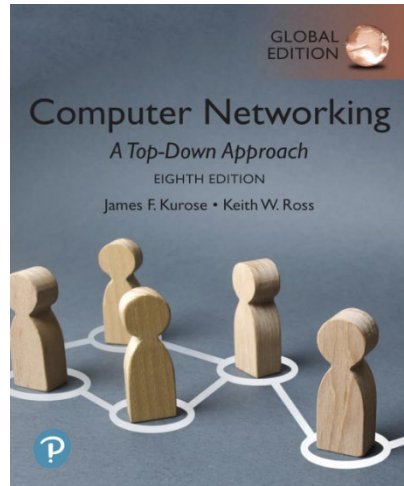
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CITS3002 Credits

- ❑ Most content is provided by Dr Chris McDonald
- ❑ Some from textbooks



Introduction: outline

- ☐ Key concepts in computer networking
- ☐ Topology
- ☐ Standards
- ☐ The Internet
- ☐ LANs & WANs

Quick stop: Terminology

Term	Meaning
Computer Network	An interconnected collection of autonomous computers
Interconnection	The capability of exchanging information. The connections can be over various medium types (e.g., copper wire, radio frequencies, optical fibre, infra-red, satellite, microwaves etc).
Autonomous	Refers to a system or computer that operates independently without a permanent master–slave relationship or centralized control from other computers.
Physical network	The actual hardware layout, including cables, switches, routers, and other networking devices.
Logical network	The virtual software-based structure showing how devices communicate over the network

Why computer network?

- ❑ Resource Sharing
 - Share hardware (printers, storage) and software among multiple users.
- ❑ Access to shared & distributed data:
 - Retrieve information stored on remote servers, databases, and cloud platforms.
- ❑ Communication & collaboration
 - Email, messaging, video meetings, and teamwork from different locations

Why computer network?

- ❑ Remote access
 - Use services and files from anywhere (mobile, home, office)
- ❑ Reliability & fault tolerance
 - Backup systems and redundancy keep services running during failures
- ❑ Cost efficiency
 - Shared infrastructure is cheaper than standalone systems
- ❑ Centralized management
 - Easier administration, security control, updates, and maintenance

Why computer network?

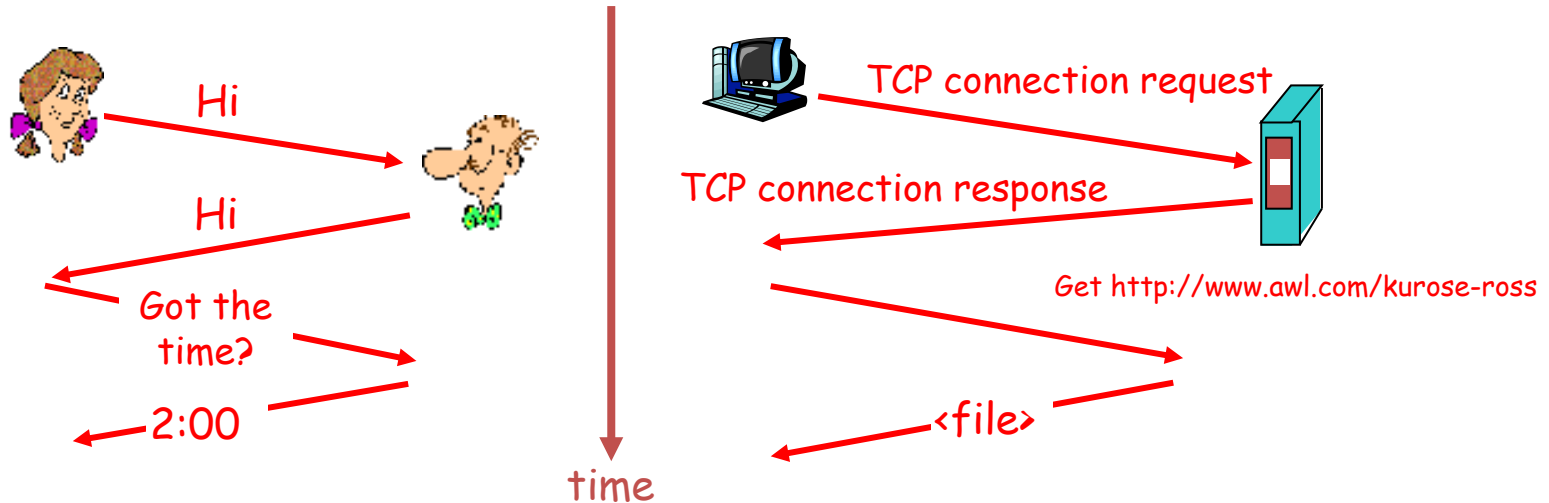
- ❑ What could be done further for computer networks?
 - Improve the reliability of the communication
 - Reduce delay for transferring data
 - Enhance security and privacy
 - Enhance parallel computing
 - Reduce energy consumption

- ❑ **Definition:** A network protocol consists of an agreed format for messages, expressed by a packet header, an optional message component, and **a set of rules for the exchange of messages between computers.**

- ❑ We see the use of protocols in computer science in almost every activity:
 - HTTP (accessing web pages and websites)
 - IPv6 (enabling devices to be connected to the Internet)
 - SSH (secure remote access over an unsecured network)
 - SFTP (secure file transfer)

Network Protocols

- ❑ An agreed set of rules that govern communication between devices
 - Both endpoints need to *understand* the protocol
 - Protocols must be formally defined and unambiguous!



Network Protocols

RED	CIDR	HTTP	RTP	RIP	ARP	FTP
IP	SNMP	HIP	ECN	BGP	SACK	TFTP
OSPF	IGMP	VLAN	POP	IPX	LDP	MAC
WAP	DHCP	UDP	TLS	ICMP	LISP	SSH
SIP	IMAP	DNS	PIM	STUN	RTSP	NAT
BFD	SMTP	VTP		LLDP	MPLS	NNTP
	RTCP	PPP			TCP	

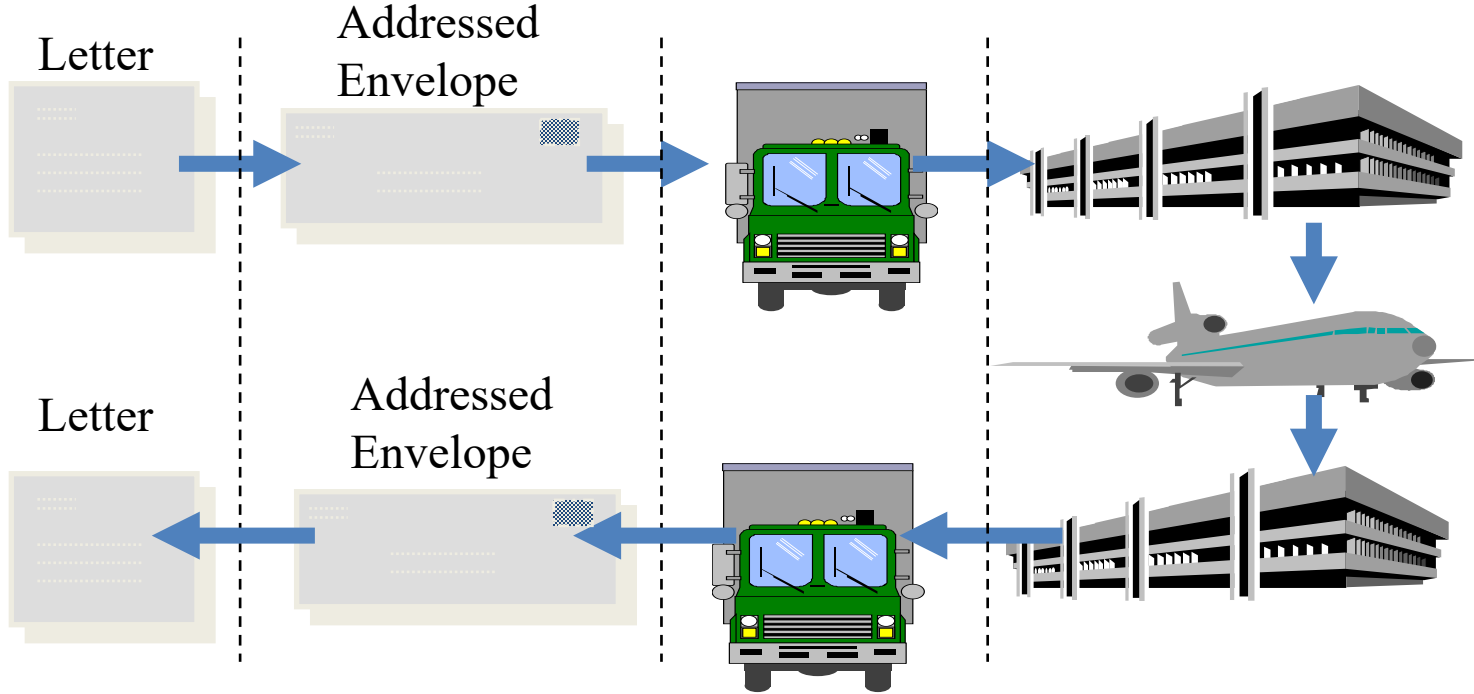
OSI Model

- ❑ With computer networks, we require protocols to "meet" new computers, ask for information, agree to share data, etc.
- ❑ The complexity of protocols can be **simplified** by separating some of the functions required into different protocols, and **isolating** layers of responsibility into different protocol modules.
 - e.g. at the lowest level we are concerned with correctly **transmitting bits** (0's and 1's) of data.
 - At another level we may be interested in transferring files between different computers which support **different data type representations**
- ❑ To manage this complexity, networking uses **a protocol stack** (e.g., the OSI model) that divides communication into layers.

Why layer?

- ❑ Each layer performs a defined set of functions within the network model.
- ❑ Mail delivery example:
 - Letter in envelope, address on outside
 - FedX guy adds addressing information, barcode.
 - Local office drives to airport and delivers to hub.
 - Sent via airplane to nearest city.
 - Delivered to right office
 - Delivered to right person

Why layer?

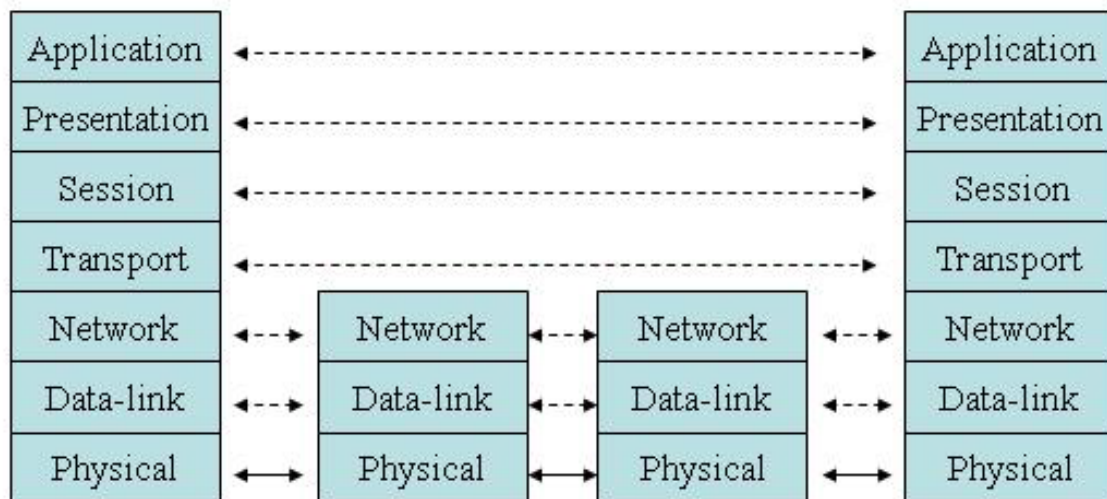


Why layer

- ❑ The following principles were followed:
 - A layer corresponds to a different level of **abstraction**.
 - Each layer provides a **well defined, independent** function.
 - Within each layer, unique protocol **standards** should be enforceable.
 - There should be **a minimum of traffic** between layers/across **interfaces**.
 - The number of layers should be **sufficiently large**
 - ❖ to separate distinct functions in different layers
 - ❖ not too many layers for the whole model to become **unmanageable**

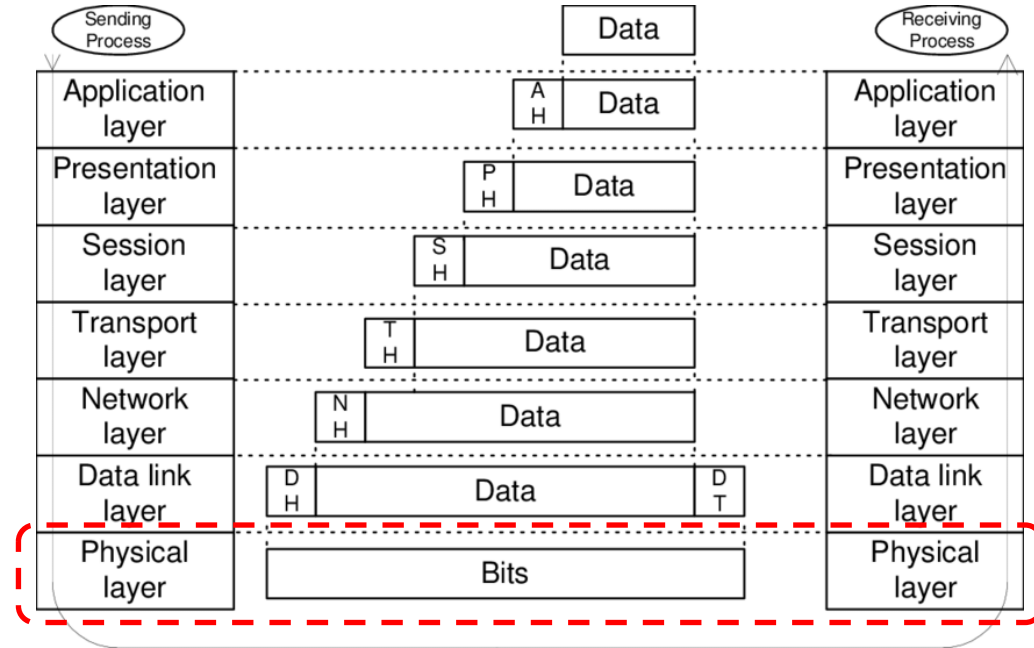
- ❑ One solution to such separation of responsibilities is provided by the ISO (International Standards Organization) OSI (Open Systems Interconnection) **reference model**.
 - (This model was finally agreed upon and standardized in 1983).
- ❑ Until the early 2000s, the OSI protocol suite was still actively sought in all tender specifications by the Australian and U.S. governments.
- ❑ However, in practice, the **TCP/IP protocol stack** became dominant and now supports most modern networks.
- ❑ While TCP/IP employs a 4-layer model, in contrast to the 7-layer ISO/OSI model, many identical concepts may be observed.

OSI Model



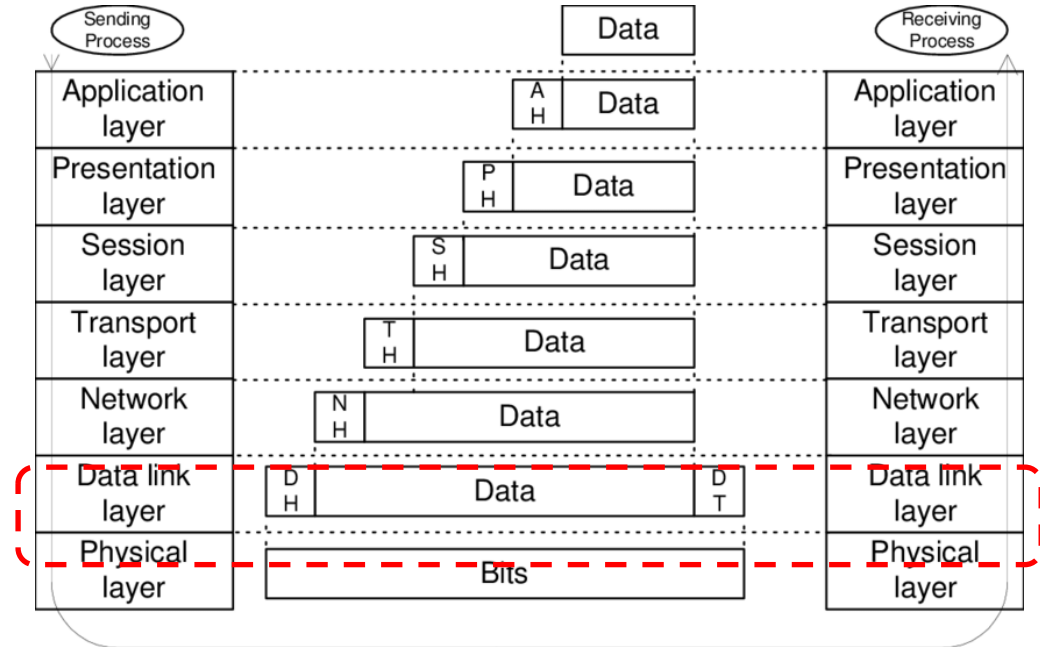
Physical Layer

- ❑ The physical layer is responsible for transmitting a **(raw) bit stream** over the physical communication medium.
- ❑ As such it is concerned with the **electrical** and **mechanical** interface between the data and the physical medium



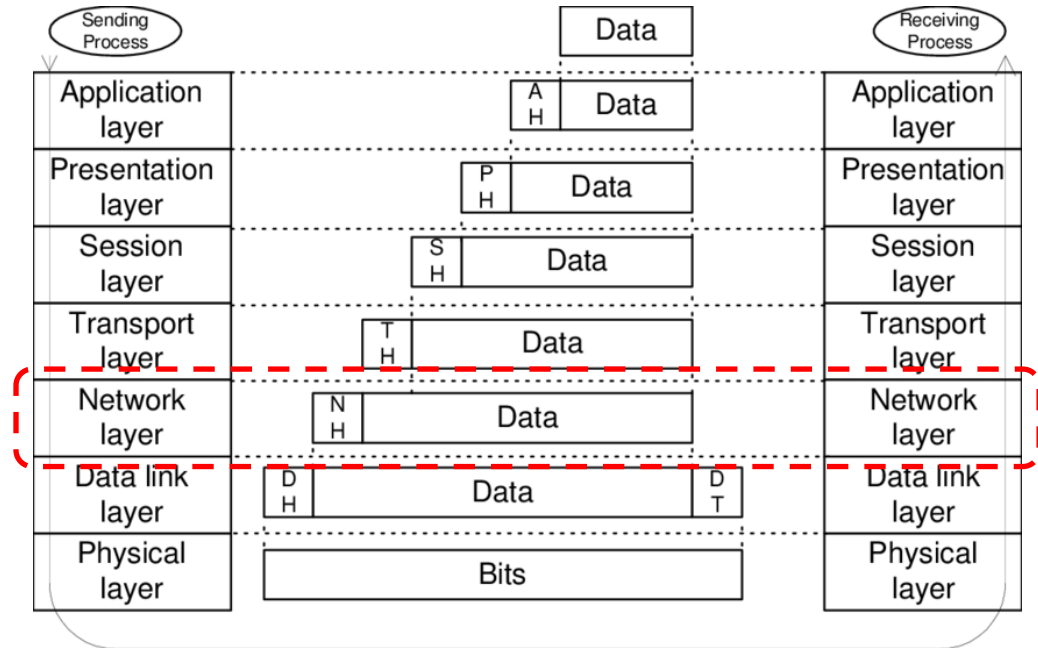
Data Link Layer

- ❑ The data link layer takes the bit stream from the physical layer and constructs **logical chunks** of data termed **frames**.
- ❑ The purpose of framing is to ensure the **reliable transmission** of information by performing limited **error detection and recovery**.



Network Layer

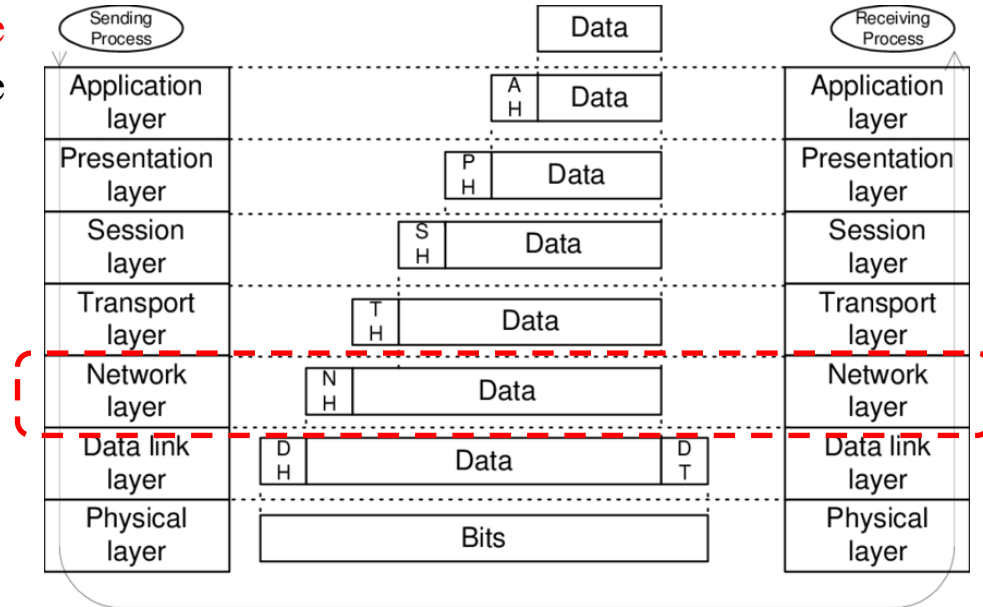
- ❑ The network layer moves data between end systems by selecting **routes** across networks using **logical addresses like IP**.
- ❑ It determines routing paths and provides **logical addressing** so communication can occur **independently** of the underlying physical networks.



Transport Layer

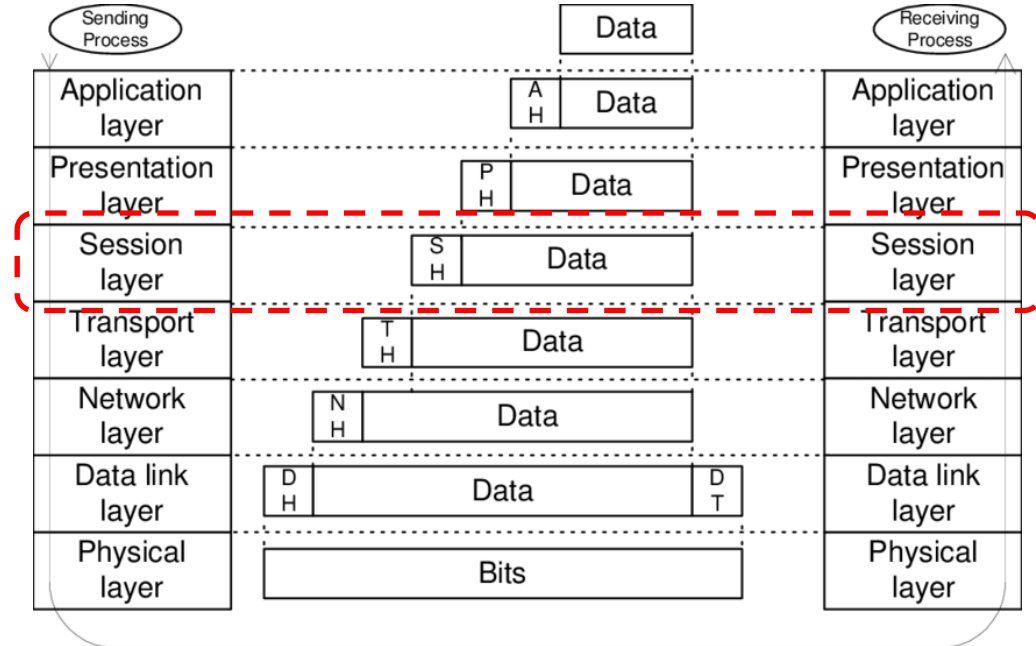
- ❑ The transport layer provides a **reliable end-to-end service independent** of the network **topology**.

- ❑ Key Functions:
 - ❑ Flow Control
 - ❑ Congestion Control
 - ❑ Connection management (TCP handshake)



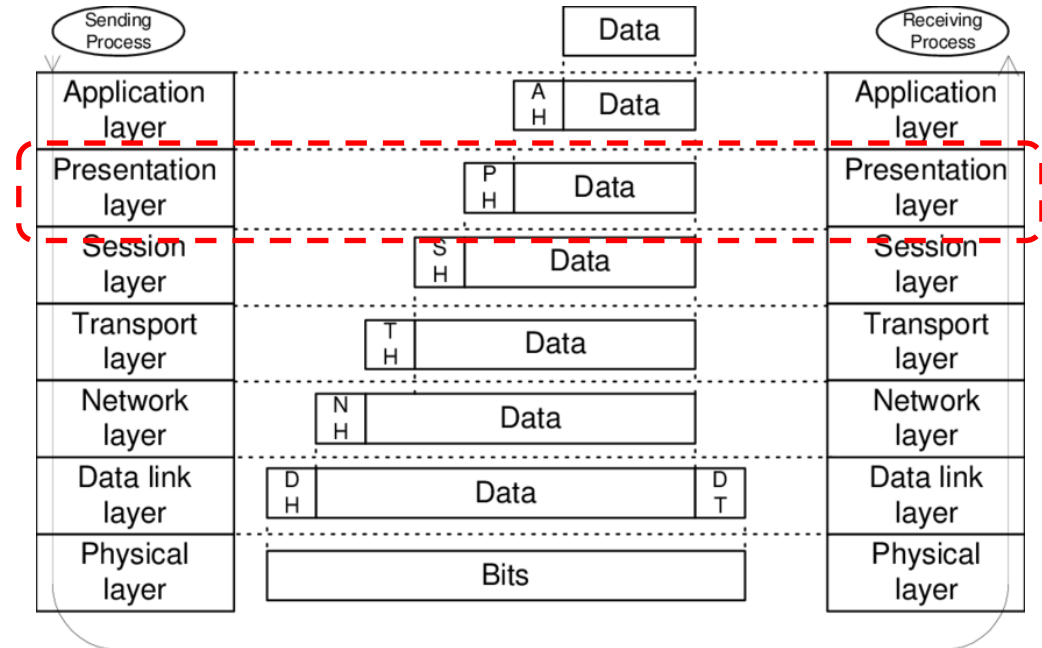
Session Layer

- ❑ The session layer is the first upper layer that manages the dialogue between end systems.
- ❑ Key operations include:
 - Manage the conversation between end users
 - Synchronization
 - Negotiation



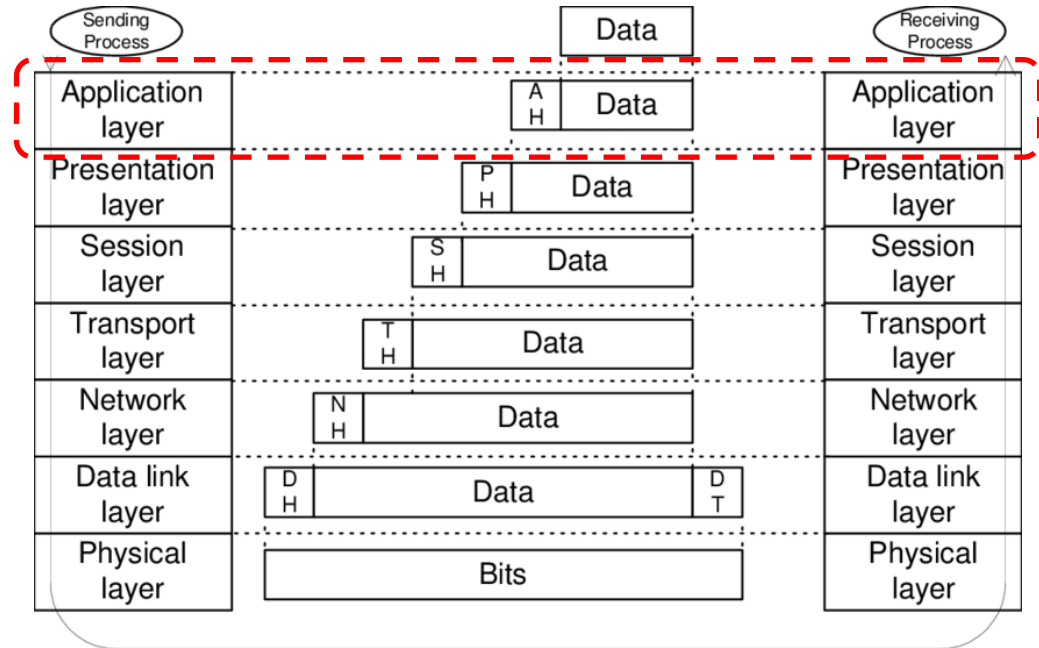
Presentation Layer

- ❑ The presentation layer provides a **standard format** for transferred information by overcoming compatibility problems between systems using **dissimilar data encoding rules** and (possibly) **different display** (input and output) technologies.



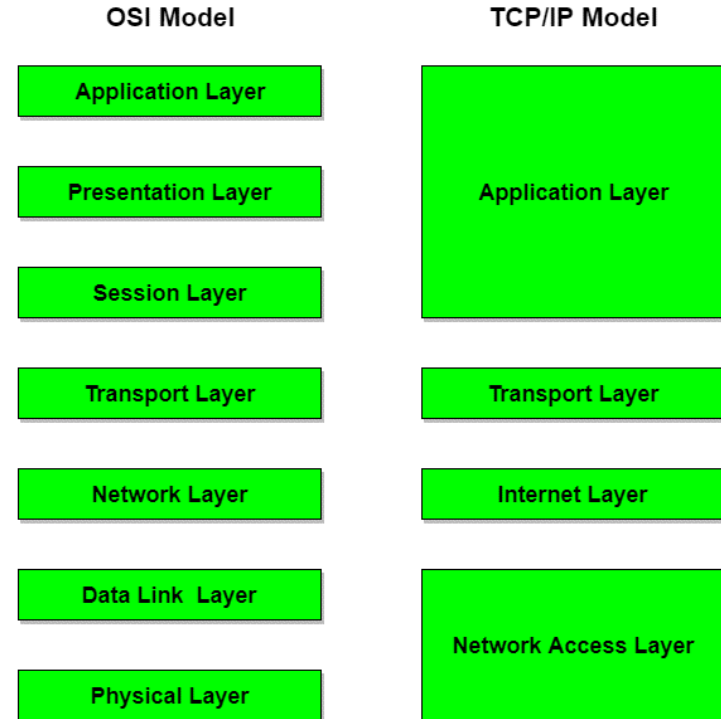
Application Layer

- ❑ Interface to user
 - Allows apps (browser, email, file transfer) to access network services
- ❑ Service request & response
 - Handles requests such as web browsing, email, file sharing
- ❑ Resource sharing
 - Enables access to remote files, printers, databases, and cloud services



OSI vs TCP/IP

- ☐ OSI model is a generic standard.
- ☐ TCP/IP model is based on the Internet.
- ☐ Both captures all aspects of current networking technologies.



Network Topologies

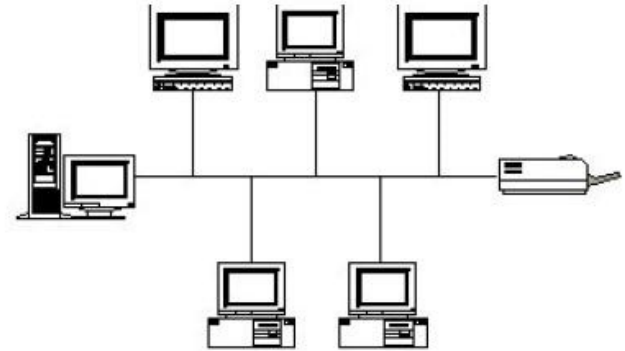
- ❑ The design of **how things are connected** can affect the **performance**.

- ❑ Topology - **Physical** and **logical** network layout:
 - ❑ Physical – actual layout of the computer cables and other network devices
 - ❑ Logical – the way in which the **network appears to the devices** that use it.

- ❑ Common topologies:
 - ❑ Bus, ring, star, mesh and wireless

Bus Topology

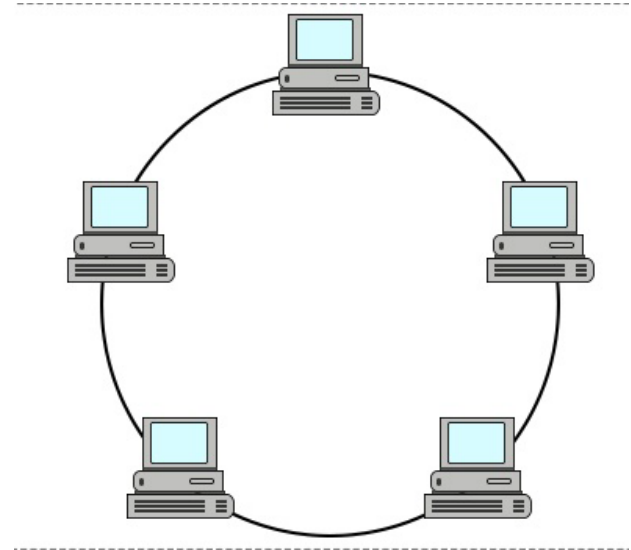
- ❑ Bus topology is a network layout where all devices connect to a **single shared communication cable** called the **backbone**.
 - All devices share **the same transmission medium**
 - Data is visible to every device on the bus



- ❑ Systems connect to this backbone using *T* connectors or taps
- ❑ If the backbone cable fails, **the entire network** stops working.

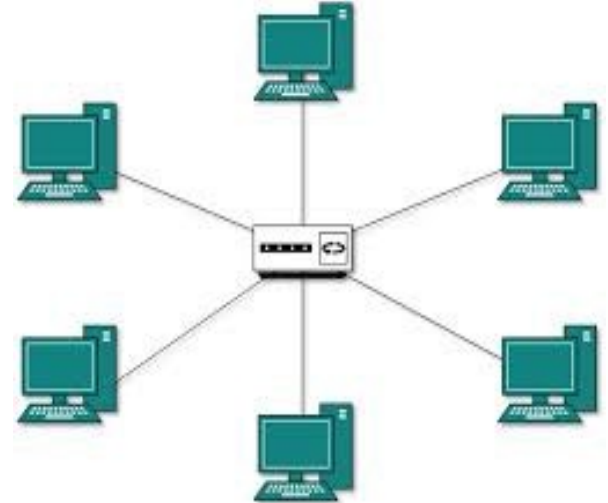
Ring Topology

- ❑ Ring topology is a network layout where each device is **connected to two neighbouring devices**, forming a closed loop (ring).
- ❑ Data travels around the ring **from one device to the next** until it reaches the destination.
- ❑ A **token** (control frame) circulates to control which device can transmit
 - Only **one sender at a time**
 - every node eventually receives the token



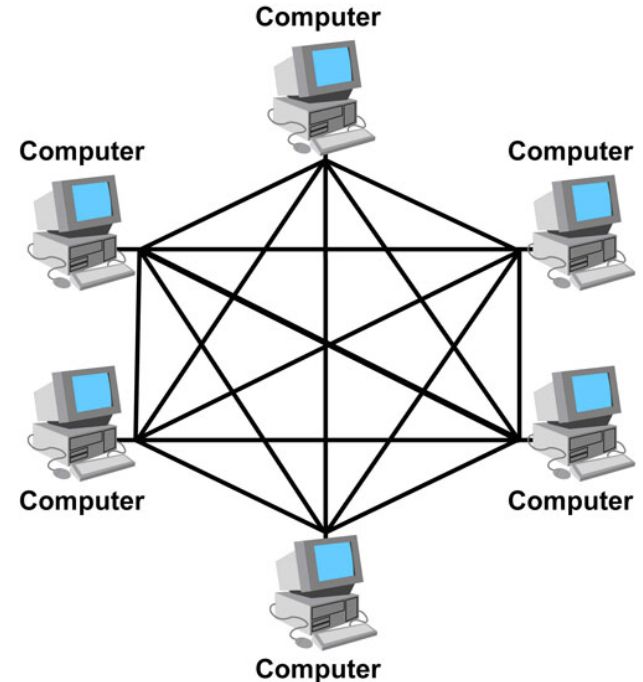
Star Topology

- ❑ All computers/devices connect to a central device called **hub or switch**
- ❑ Each device requires **a single cable**
- ❑ Point-to-point connection between the device and the hub
- ❑ **Most widely implemented**
- ❑ Hub is the single point of failure



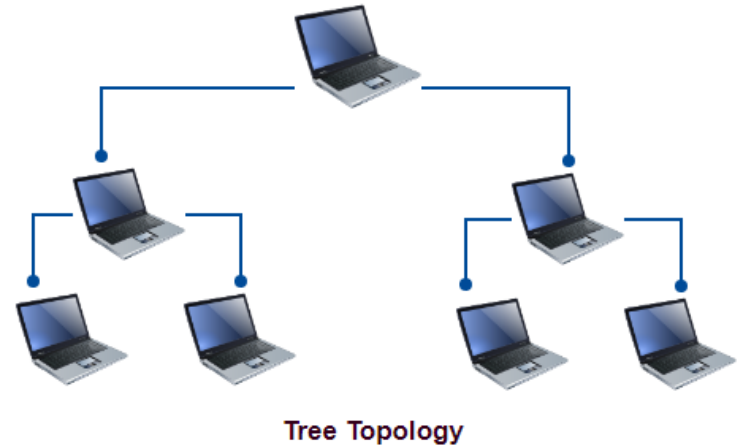
Mesh Topology

- ❑ Each computer connects to every other (or many other).
- ❑ High level of **redundancy**
- ❑ Rarely used:
 - Wiring is very complicated
 - **Cabling cost** is high
 - Troubleshooting a failed cable is tricky
 - Adding a new node requires **multiple new connections** in full mesh



Tree Topology

- ❑ In computer networks, a tree topology is also known as a star bus topology.
- ❑ Typically has a root node, branching out to children nodes that connects those hosts together.
- ❑ Each branch can be seen as a star topology connected to a backbone
- ❑ Easy to expand by adding new branches



Wireless Networks

- ❑ Do not require physical cabling
- ❑ Eliminate cable faults and cable breaks
- ❑ Signal interference and security issue



Network Standards: IEEE

- ❑ Institute of Electrical and Electronics Engineers (IEEE) develops widely adopted networking standards
 - Focus: hardware and lower-layer networking technologies

- ❑ Networking technologies from different manufacturers are designed to be interoperable

- ❑ IEEE standards ensure compatibility across cabling, networking devices, and communication protocols

IEEE Standards

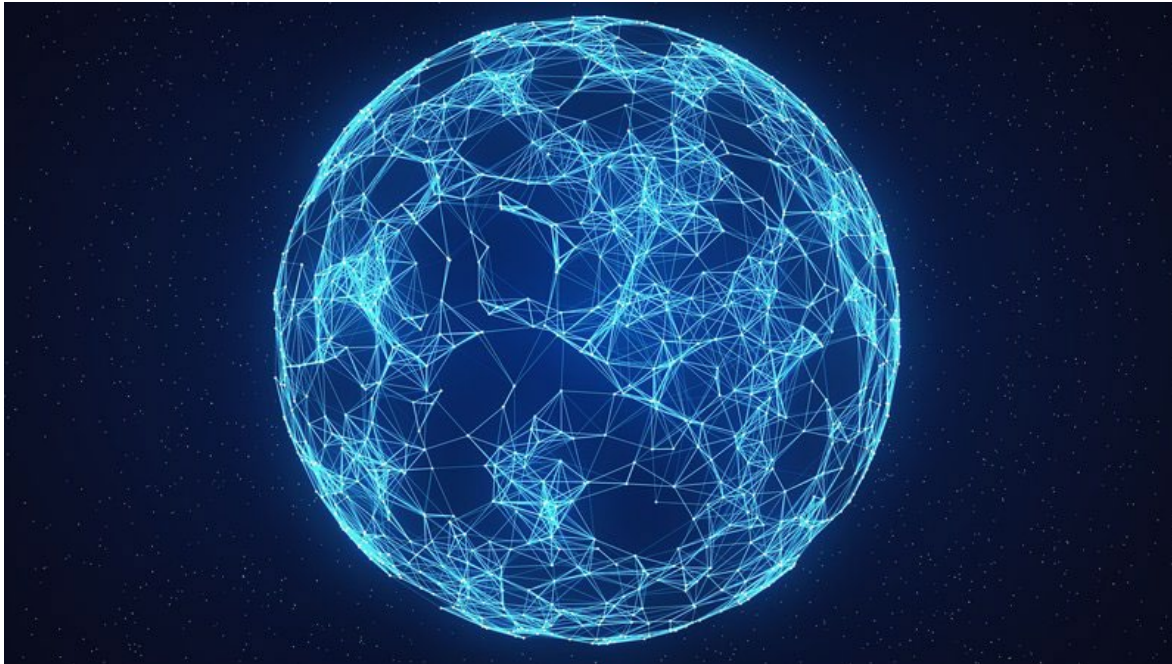
Specification	Name
802.1	Internetworking
802.2	The LLC (Logical Link Control) sublayer
802.3	CSMA/CD for Ethernet networks
802.4	A token passing bus
802.5	Token Ring network
802.6	Metropolitan Area Network (MAN)
802.7	Broadband Technical Advisory Group
802.8	Fibre-Optic Technical Advisory Group
802.9	Integrated Voice and Data Networks
802.10	Standards for Interoperable LAN/MAN Security (SILS) (Network Security)
802.11	Wireless networks
802.12	100Mbps technologies

Network Standards: IETF

- ❑ IETF (Internet Engineering Task Force) is an international organization that develops and promotes Internet protocol standards to ensure interoperability across networks.

- ❑ Key characteristics:
 - Open, global community of engineers and researchers
 - Focuses on Internet architecture and protocols
 - Standards published as **RFCs** (Request for Comments)
 - ❖ RFC 8200 - Internet Protocol (IPv6)
 - ❖ RFC 793 - TCP (Transmission Control Protocol)
 - Consensus-based and collaborative process

The Internet



The Internet - history

- ❑ Internet:
 - One of the most transformative inventions of modern society
 - Originated as a research experiment (ARPANET) and evolved into the global communication infrastructure
 - Rapid and continuous expansion in scale and impact

- ❑ Global reach:
 - Today: 6+ billion Internet users worldwide
 - Massive growth in connected devices (IoT, mobile, cloud systems)

- ❑ Applications: Web, cloud computing, social media, streaming, P2P, immersive/virtual environments

The Internet - influence

- ❑ The ways we do business:
 - E-commerce, advertising, cloud computing, ...
- ❑ The way we have relationships
 - E-mail, IM, Facebook, Instagram, virtual worlds
- ❑ How we think about law
 - Interstate commerce? National boundaries? Wiki-leaks?
- ❑ The way we govern
 - E-voting
 - Censorship and wiretapping
- ❑ The way we fight
 - Cyber-attacks, nation-state attacks, fake news

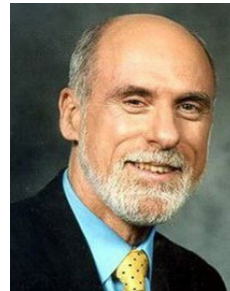
The Internet

- ❑ The term “Internet” simply refers to the *network of networks*.
- ❑ The protocols allow **different** networks to communicate to one another.
- ❑ Is Internet the same as www (world wide web)?
 - **No.** www is a system that runs on top of the Internet (currently rely mostly on HTTP), while the Internet is an infrastructure that supports many other functionalities.

The Internet

❑ Who made it?

- Notably, the ARPANET project in the early 1960s
- TCP/IP was developed by *Robert Kahn* and *Vint Cerf* in the 1970s, which became the standard protocol for the ARPANET, which was decommissioned later in 1990.



The Internet

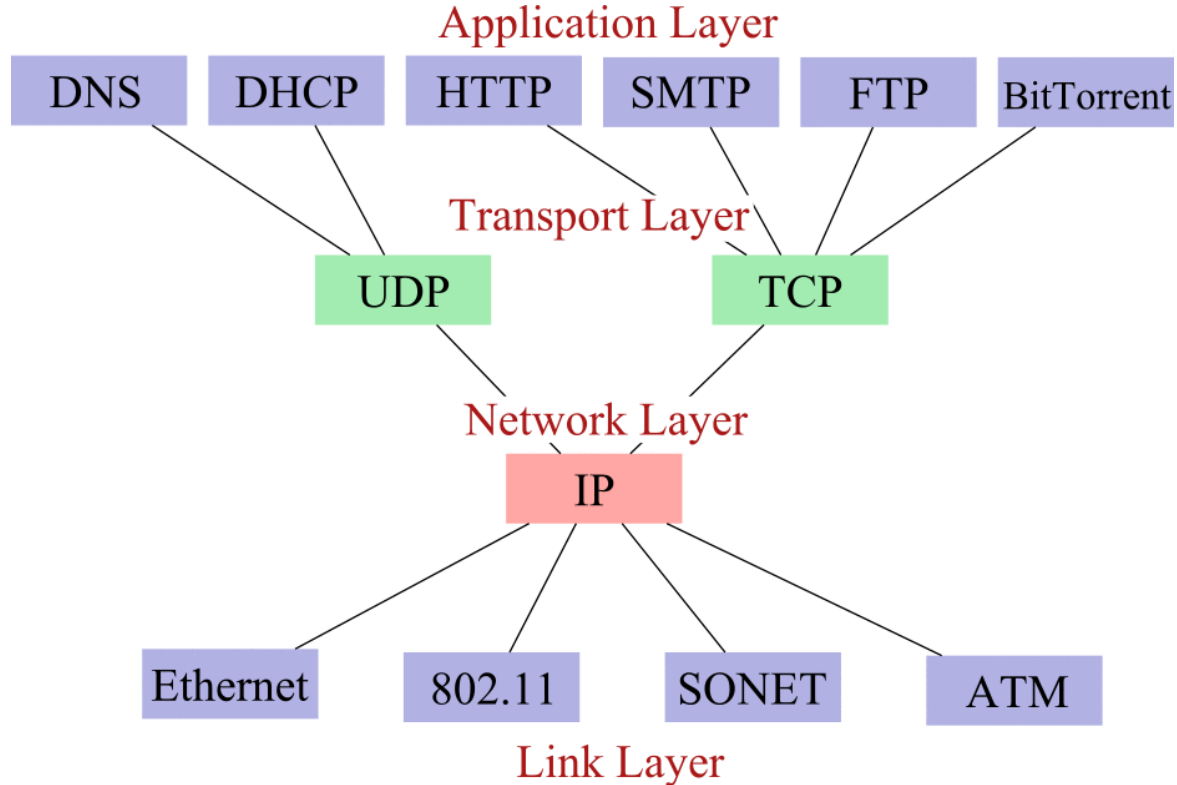
- ❑ The main goal was the **interoperability** among existing networks
 - a network of networks
 - obey administrative boundaries
- ❑ Additionally, we also wanted to have
 - fault tolerance
 - multiple transport protocols
 - support a variety of networks

Lots of attention

 - distributed management
 - cost effective
 - low effort for host attachment
 - accountability

Less attention...

The Internet



The Internet

❑ The underlying infrastructure varies

❑ Wired

- Coaxial
- ITU-T G.hn (coaxial, phone and power lines)
- Twisted pair cabling (e.g., Ethernet)
- Optical fibre



❑ Wireless

- Microwave
- Satellite
- Cellular



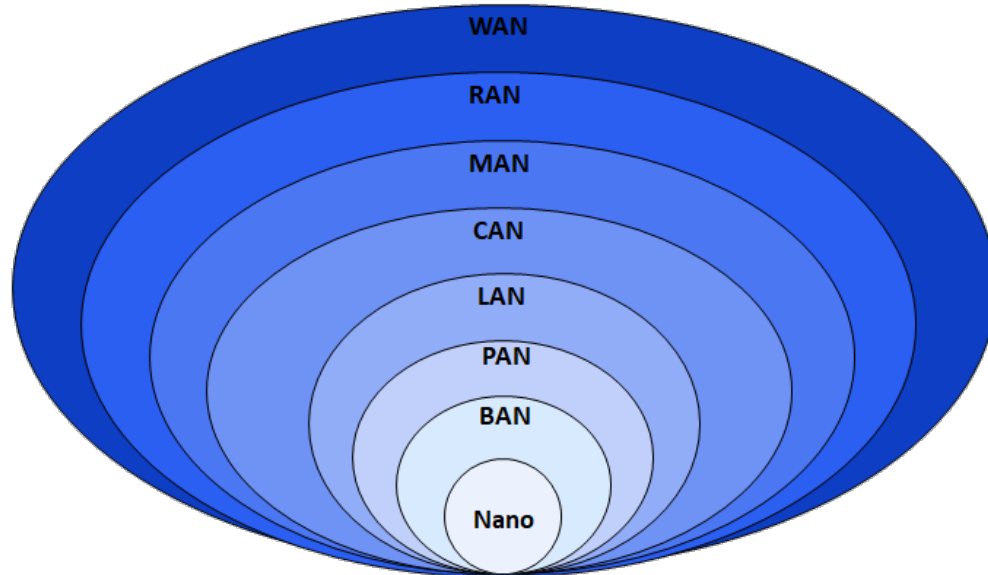
The Internet

❑ Challenges in networking

- Throughput
- Latency
- Reliability
- Cost
- ...

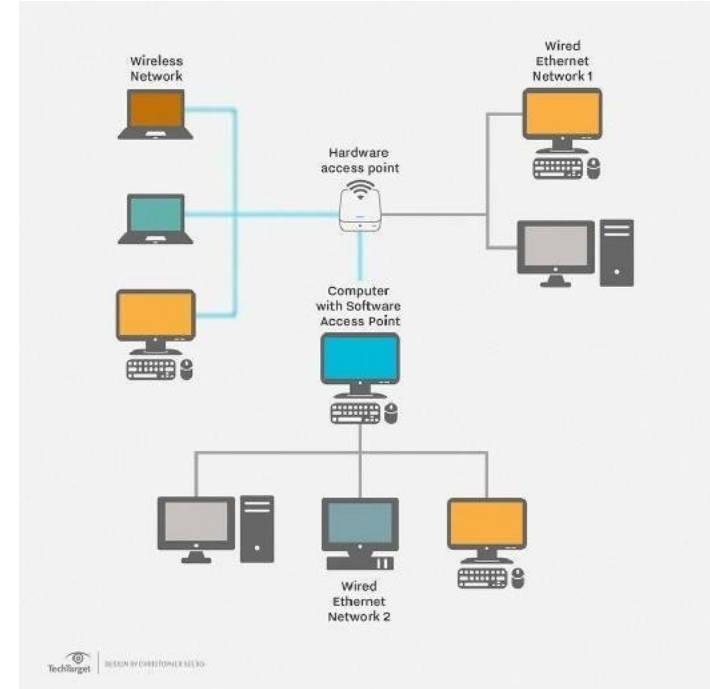
The Network

- ❑ There are many different types of networks:
 - LAN, WAN, WLAN, MAN, CAN, BAN etc...



LANs

- ❑ Physically close devices connected together
 - E.g., home, business, enterprise etc.
- ❑ Typically provides
 - high speed
 - multi-access
- ❑ Technologies:
 - Ethernet



- ❑ Connects computers that are physically far apart.
 - a high-capacity communication network that connects cities, countries, or continents over long distances
 - Typically slower and less transfer rate than LAN
 - Point-to-point
- ❑ Technologies:
 - ATM
 - Fiber optics
 - Cable
 - Wireless WAN

